Describe a wave

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify amplitude, wavelength, frequency and period.

SUCCESS CRITERIA - I can...

1. Measure amplitude from equilibrium.
2. Measure wavelength between matching points.
3. Connect frequency and period.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A sample warms by 3°C after a certain energy input. If absorbed energy becomes 5 times as large with the same mass and material and no phase change, find the new rise.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: f = cycles/Δt; T = 1/f.
f is frequency (Hz); T is period (s). λ is crest-to-crest wavelength; amplitude is maximum displacement.

Worked reasoning: A wave makes 6 cycles in 3 s: f=2 Hz and period T=0.5 s. Amplitude measures maximum displacement, not the spacing of crests.

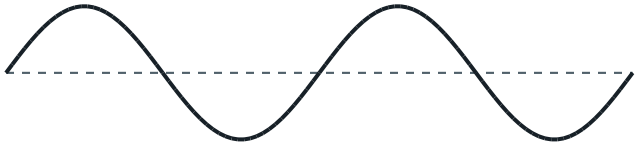
Your turn: Draw one spatial wave and label amplitude and wavelength 0. Separately mark one period on a displacement-time graph.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Label amplitude and one wavelength on the wave diagram. Explain why crest-to-trough height is not the amplitude.



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A point on a rope completes 6 cycles in 5 s. Find its frequency and period. In a separate trial, each cycle takes 0.5 s. Find that trial's frequency. Use two time traces to show which trial oscillates more often.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Does increasing amplitude necessarily increase frequency? Use the definitions to explain 0.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg sample has c=600 J/(kg·°C) and warms by 8°C without changing phase. Find Q.
Label temperatures, samples and relevant energy transfers.

Sample A

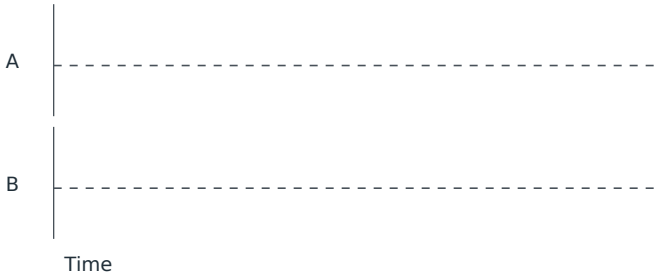
Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

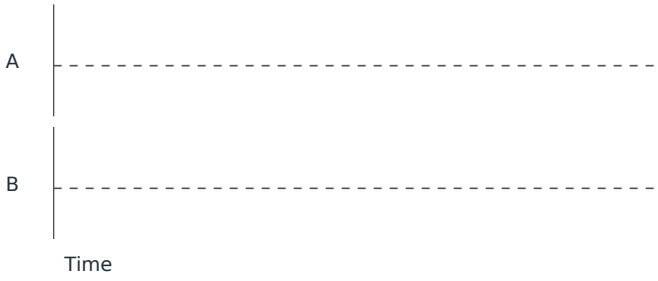
A wave has period 2 s. How many cycles occur in 6 s? Find frequency.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Draw two waves with equal wavelength but different amplitudes. Explain which feature changes and which does not.



Marzano check - circle: 1 2 3